

7 (a) (i) Solution 1

$$A_0 = \lambda N_0$$

$$\begin{aligned} &= \left(\frac{\ln 2}{50.2 \times 60 \times 60} \right) \left(\frac{2.00 \times 10^{-9}}{240 \times 10^{-3}} \times (6.02 \times 10^{23}) \right) \\ &= 1.9241 \times 10^{10} = 1.92 \times 10^{10} \text{ Bq} \end{aligned}$$

Solution 2

$$A_0 = \lambda N_0$$

$$\begin{aligned} &= \left(\frac{\ln 2}{50.2 \times 60 \times 60} \right) \left(\frac{2.00 \times 10^{-9}}{240 \times 1.66 \times 10^{-27}} \right) \\ &= 1.9254 \times 10^{10} = 1.93 \times 10^{10} \text{ Bq} \end{aligned}$$

(ii) number of decays required = $\frac{1140}{(5.71 \times 10^6)(1.60 \times 10^{-19})}$

$$= 1.2478 \times 10^{15} = 1.25 \times 10^{15}$$

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(iii) Solution 1

$$\begin{aligned}
 N_{\text{required}} &= N_i - N_f \\
 &= N_0 - N_0 e^{-\lambda t} \\
 1.25 \times 10^{15} &= \left(\frac{2.00 \times 10^{-9}}{240 \times 10^{-3}} \times (6.02 \times 10^{23}) \right) \left(1 - e^{-\frac{\ln 2}{50.2} t} \right) \\
 0.249 &= 1 - e^{-\frac{\ln 2}{50.2} t} \\
 t &= -\frac{50.2}{\ln 2} \times \ln 0.751 \\
 &= 20.738 = 20.7 \text{ h}
 \end{aligned}$$

Solution 2

$$\begin{aligned}
 N_{\text{required}} &= N_i - N_f \\
 &= N_0 - N_0 e^{-\lambda t} \\
 1.25 \times 10^{15} &= \left(\frac{2.00 \times 10^{-9}}{240 \times 1.66 \times 10^{-27}} \right) \left(1 - e^{-\frac{\ln 2}{50.2} t} \right) \\
 0.249 &= 1 - e^{-\frac{\ln 2}{50.2} t} \\
 t &= -\frac{50.2}{\ln 2} \times \ln 0.751 \\
 &= 20.738 = 20.7 \text{ h}
 \end{aligned}$$

- (b)** The statement is incorrect because the activity of a radioactive sample is also proportional to the number of radioactive nuclei (besides being inversely proportional to its half-life).

So radioactive materials with a long half-life can have high activity if the number of radioactive nuclei is large.